

Problem

The time constant for an RL circuit with $R = 2\ \Omega$ and $L = 4\ \text{H}$ is:

- (a) 0.5 s
- (b) 2 s
- (c) 4 s
- (d) 8 s
- (e) 15 s

Step-by-step solution

Step 1 of 1

Calculate the time constant for an RL circuit using the following formula.

$$\tau = \frac{L}{R}$$

Substitute $2\ \Omega$ for resistance R and $4\ \text{H}$ for inductance L .

$$\begin{aligned}\tau &= \frac{4}{2} \\ &= 2\ \text{s}\end{aligned}$$

Thus, the value of the time constant for an RL circuit is $2\ \text{s}$. Hence the correct option is **b**.

As the option b is correct remaining options are not suitable for the correct option.